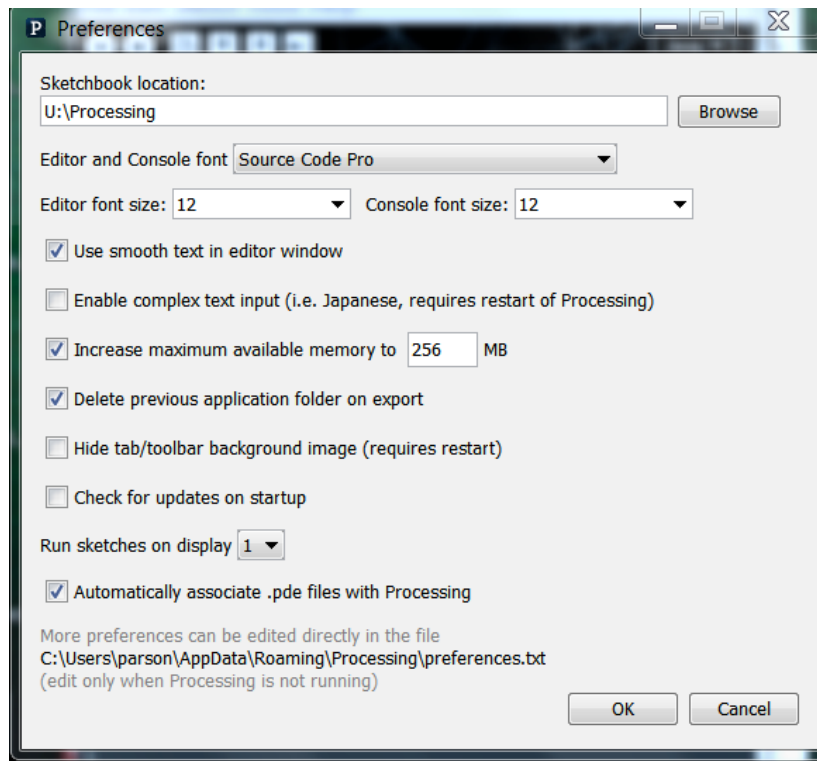


CSC 120 Introduction to Creative Graphical Coding, Spring 2021

Assignment 1, Implementing and testing an automated avatar in Processing.

This assignment is due via **D2L Assignments Assignment 1 avatar** by **Feb. 20, 2021**

When using Processing on the Kutztown campus Windows computers, make sure to start out **every time** by setting your Processing Preferences -> Sketchbook Location to S:\Processing. The S:\ drive is a networked drive that will save your work and make it accessible across campus. If you save it to your desktop or the lab PC you are using, you will lose your work when you log out. You must save it to the S:\ drive. If you do not have a folder¹ called Processing under S:\, you must create one using the Windows Explorer. Processing Preferences is under the File menu on Windows.



If you will be downloading Processing 3.X and running it using an off-campus computer (do not use version 2.X for assignments), you can copy your project sketch named **avatar** to a flash drive on one machine, and then copy it from the flash drive to another Processing sketch folder.

Create your **avatar** folder by running File -> Save As -> **avatar** after setting up your sketch folder.

Once you have created your **avatar** sketch, write the following code as outlined in class:

1. Create some immobile background scenery. (Hint: Plot this before the avatar.)
Keep them opaque, differing colors.

¹ Another name for a folder is a *directory*.

2. Create an avatar that is **not** the textbook's Zoog, sample avatar, or another classmate's avatar. Call **pushMatrix()** before starting the avatar to save the window based x,y coordinate system. Call **translate(...)** with coordinates for the center of the avatar's body. 0,0 is now its reference point; use the center of its body for this approximate reference point. Optional bonus points 1.5 for using **scale**, another 1.5 for using **rotate** for the avatar. Use at least one shape from "2D Primitives" (on the manual page) that Zoog does not use. Zoog uses ellipse, line, and rect. Note use of rectMode and ellipseMode; I recommend CENTER. Use variations in color. Use variations in alpha. Use at least 5 distinct shapes, meaning at least 5 body parts for the avatar. Call **popMatrix()** before resuming display of foreground scenery; it restores 0,0 to top left.
3. Create some immobile foreground scenery. (Hint: Plot this after the avatar.) Keep them opaque, varying colors across time.
4. Your avatar must move horizontally and/or vertically at some rate.
5. Your avatar MUST wrap back around to the other side of the window, OR bounce back in opposite, when hits window's edge.

See sample code in D2L on how to make your avatar return or reappear.

6. Your avatar must have some "body part" that wiggles, grows/shrinks, or moves in some manner, without becoming disconnected from the avatar. Step 6 may use the modulo operator (%), which gives the remainder of division, to wrap back around to its starting point. You may use an "if" statement instead.

Use colors and alpha (stroke() and fill() and background()), strokeWeight(), and a variety of shapes. Get into creating a composition. Consult <https://processing.org/reference/> and use some functions to make your animation exciting.

Grading: Each of the above 6 steps is worth 15% each (90% total). I will give half credit for step 4 if wrap-around does not work.

That leaves 10% for compositional effort, by which I mean creating a scene that is a composition, as opposed to just a random collection of graphical features that satisfy the above requirements. Students have done campus scenes, city scenes, and scenes in space with a flying avatar. Create an actual scene.

Don't forget about the automatic 10% penalty to assignments that don't follow the CS&IT department's standards for documentation. For this assignment, that just means placing this comment block at the top of your code:

```

/*****
* Author:
* Creation Date:
* Due Date:
* Course:
* Professor Name:
* Assignment:
* Sketch name:
* Purpose:
*****/

```

Also, within your `setup()` and `draw()` functions, add some comments using the “//” comment delimiter to describe the avatar body part you are plotting, similar to how the textbook author describes Zoog.

We will have some time in class for working on this project. If you do not get it done in class, you will have to complete it as homework.

TURNING IT IN: Upload your completed file to the D2L assignment dropbox. Remember you have 3 budgeted “grace-days” for this semester. See the *Assignments/Projects* section of my first day handout for more information.